Beta Parameter Equation for NTC Thermistors

The relationship between the resistance and temperature of an NTC thermistor is originally expressed as:

$$R_T = R_0 \cdot e^{\beta\left(\frac{1}{T} - \frac{1}{T_0}\right)}$$

which was transformed to:

$$T = \frac{1}{\frac{1}{T_0} + \frac{1}{\beta} \ln\left(\frac{R_T}{R_0}\right)}$$

Where:

- R_{T}: The resistance of the thermistor at the current temperature.
- R_0: The nominal resistance at room temperature (for an NTCC-10K, this is 10 kΩ).
- T_0: The nominal room temperature in Kelvin (25°C = 298.15 K).
- T: The current temperature in Kelvin.
- β: The Beta constant of your specific thermistor. It equals $4050 \pm 10\%$ K according to the datasheet of the NTCC-10K thermistor.